

M.R.R. Project 2025

Methodology Report on breast cancer survival prediction

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1 Introduction

Our primary objective is focused on **finding the genes expressions that have a significant impact on the breast cancer survival outcome of patients.**

2 Data pre-processing

During data processing, the first essential step was to change the type of all non-numeric columns (which were lists of characters, characters, etc.), such as clinical variables or patient identifiers, into the factor type. This ensures that all our functions correctly process the data for these variables.

Since there was no missing data, we proceeded directly to handle categorical sparsity. We identified and removed too rare levels (levels with very few observations) from these factor variables using custom thresholds. This is essential to prevent these low-frequency levels from negatively affecting model stability and training.

Once the data types were corrected and rare levels were appropriately handled, we set aside variables and observations that were objectively poor in information, which include variables with near-zero variance.

2.1 Gene Expression Clustering

As a complementary exploratory step to data pre-processing, we performed an unsupervised **k-means** clustering analysis on the gene expression variables to identify potential molecular subgroups of breast cancer patients. Based on internal validation criteria such as within-cluster inertia and silhouette scores, **four clusters** were selected as the optimal partition. Patients were then assigned to one of these clusters using their gene expression profiles.

To assess the biological relevance of this unsupervised structure, we compared our four-cluster solution with the **PAM50 molecular classification** (LumA, LumB, HER2-enriched, Basal-like, Normal-like). The results exhibit a **strong overall agreement** between both approaches, with clear correspondences for the Basal-like, HER2-enriched, and Normal-like subtypes.

The main discrepancy concerns the **Luminal A and Luminal B** subtypes, which tend to be grouped together in our clustering. This outcome is expected given the strong molecular similarity between these two subtypes.

3 Data splitting and model evaluation

We then **split our refined dataset into two parts** :

- One part to train our different models ($\sim 70\%$ of the dataset).
- The other part to test the effectiveness of these models ($\sim 30\%$ of the dataset).

Since the survival outcome is a binary variable (e.g., survival vs. non-survival), our task is a classification problem. To compare the effectiveness of the different models, we will build confusion matrices using the test data and compute standard performance metrics such as accuracy, precision, recall, and F1-score. To convert predicted probabilities into classes, we will use an optimized threshold determined on a validation set, rather than the default 0.5 cutoff because of the strong imbalance in the dataset.

3.1 Problem : the strong imbalance in the dataset

To address the strong class imbalance in the original dataset (12% deceased), we compared two alternative and independent strategies. First, the training data was resampled using **upsampling** in order to create a balanced training set with equal representation of both classes. In this setting, no class weights were applied, as the imbalance is already corrected at the data level. The test set was left untouched to preserve the original class distribution, ensuring an unbiased evaluation of model performance under real-world conditions.

Second, instead of resampling, we implemented a **class-weighted learning** approach on the original imbalanced training data by assigning a higher misclassification cost to deceased patients (since they represent a minority). This cost-sensitive strategy allows the model to account for class imbalance directly through the loss function without modifying the data distribution.

3.2 Model evaluation

Eventually, we may perform **K-fold cross-validation** using the `caret` library to obtain more robust and unbiased performance estimates. Special attention will be paid to rare categorical levels, as increasing the number of folds reduces the size of each training subset, which can lead to certain factor levels being absent in some folds and potentially destabilizing the model. To mitigate this, we may employ **stratified folds** to preserve the outcome distribution, and explore weighting or repeated CV strategies to ensure stable and reliable evaluation.

4 Simple models

First, a natural approach consists in discarding the variables that are not statistically significant for the linear regression model. This approach is particularly relevant, and we studied in the course three variable selection methods commonly referred to as “greedy” algorithms : forward selection, backward elimination, and stepwise regression.

5 Advanced Models (penalized)

To improve model performance and handle high-dimensional gene expression data, we will implement penalized logistic regression models, including **Ridge**, **Lasso**, and **Elastic Net**. These models allow regularization to prevent overfitting and perform variable selection in the case of Lasso and Elastic Net.

5.1 Ridge, Lasso and Elastic-Net regression

We will use the `glmnet` package in R to fit Ridge ($\alpha = 0$) and Lasso ($\alpha = 1$) logistic regression models. Hyperparameters (λ) will be selected using 10-fold cross-validation. For Ridge, Lasso and Elastic-Net, the models will be evaluated at both $\lambda_{\min}$ and λ_{1SE} to explore the trade-off between model complexity and stability. To combine the benefits of Ridge and Lasso, we will perform an **Elastic Net** regression. An α grid from 0 to 1 will be tested with 10-fold cross-validation to find the optimal combination of L1 and L2 penalties. The selected α and corresponding λ minimizing the cross-validated deviance.

5.2 Custom Lasso

Given the challenges of high-dimensional data with correlated features and a strong class imbalance (12% deceased), we may explore advanced Lasso-based methods beyond standard Lasso and Elastic Net to improve predictive performance and interpretability.

One approach we may consider is **Accept-Reject Lasso (ARL)**, which handles correlated features by differentiating truly redundant from spurious ones, potentially improving feature selection stability and F1-score. **Stability Selection via Variable Decorrelation** may further enhance consistency by reducing predictor correlation, while **Univariate-Guided Sparse Regression (uniLasso)** may yield sparser, more interpretable models that respect univariate-multivariate coefficient signs.

Additionally, we may consider **forcing the variable representing cancer subtypes** (obtained from gene expression clustering) to be included in all models, ensuring that this biologically meaningful factor is always accounted for while Lasso selects the remaining predictors.